

IAL Biology Unit 3 (WBI13): Rapid Revision

Practical Skills in Biology I · last-minute summary · 1 h 20 min · 50 marks · 3 questions of about 16 marks each, no mark ramp · 1.6 min per mark, the most generous paper in the qualification

HOW THE MARKS WORK

- **In all 21 papers:** graphs, calculations, and describing a method.
- **You never compute a formal statistical test.** The *t*-test appears in 0 of 21 papers and correlation coefficients in 0. But “significant” appears 37 times in the mark schemes — you **interpret standard deviations and error bars** instead.
- Most extended answers are marked as a **list of independent points**. Write short, separate, numbered sentences.
- **Quantities, not “amount”. Stated numbers**, not “several”. Say **how** a variable is controlled. Make every effect **directional**.

THE GRAPH MARKS

Graphs are marked on five codes. Learn them as a checklist.

Code	What the examiner checks
A	Axes correct way round — independent variable on the <i>x</i> axis — and scales that are linear and use at least half the grid in both directions
L	Labels with units , in the form “time / s”. A missing unit loses this
P	Plotting accurate to within half a small square
S	The right line or chart type : bar chart for discrete categories, line graph for a continuous IV, scatter for a relationship. Never dot-to-dot when a best-fit line is wanted
K	A key when more than one data set is plotted

Bar chart = discrete categories, bars separated. **Histogram** = continuous data in classes, bars touching. **Line graph** = continuous IV. **Scatter graph** = looking for a relationship between two continuous variables.

Rate from a graph: draw a **tangent**, take the gradient, give **units**. Show the construction triangle on the graph itself.

DESCRIBING A METHOD

THE UNIVERSAL MARK POINTS

Write one sentence each; every separate correct point is a mark.

1. **DV with how it is measured** — “the **volume** of oxygen released **per minute**”, not “the amount of gas”.
2. **IV with a range of stated values** — “0, 10, 20, 30 and 40 °C”, not “different temperatures”.
3. How the IV is changed.
4. The **apparatus** used to measure the DV.
5. **Standardise the material** — same mass, length, age,

source, variety.

6. **Equilibrate or acclimatise** before measuring.
7. A **stated time** for each measurement.
8. **Calculate a rate**.
9. **Two controlled variables and how each is controlled**.
10. A **control** for comparison.
11. **Repeats, then a mean and a standard deviation**.

CONTROLLING VARIABLES

Variable	Control method the schemes accept
Temperature (solution)	Thermostatic water bath or incubator — “water bath” unqualified is ignored
Temperature (aquatic plant)	Thermostatic water bath plus a heat shield between lamp and plant
pH	Buffer at a stated pH
Light intensity	Same bulb at a fixed, measured distance ; or a light meter
Mass or length of tissue	Balance; ruler; cork borer for equal diameter — “size” is ignored
Age or variety	Same batch; same cultivar; planted at the same time
Volume of solution	Measuring cylinder or pipette, stated volume

THE EFFECT MUST BE DIRECTIONAL

“It would affect the results” scores **nothing**. Write: “a higher temperature would increase enzyme activity, **so more product would form**”. The safe general fallback is “**the results would not be valid**” — never “not reliable” or “not accurate”.

THE RECURRING PRACTICAL CONTEXTS

Context	What to watch for
Plant tissue in sucrose or salt (12/21)	Potato or beetroot cylinders; measure change in mass or length , express as % change ; blot before weighing; the point of equal mass gives the water potential of the tissue
Enzymes (6/21)	Control pH with a buffer and temperature with a thermostatic bath; measure product formed per unit time
Germination (6/21)	Standardise seed batch, soaking time and depth; count or measure after a stated time
Graticule and microscope (6/21)	Calibrate the eyepiece graticule against a stage micrometer at that objective; measure a stated number of cells and take a mean

Beetroot and membrane permeability: pigment leaks because higher temperature or solvent **disrupts the phospholipid bilayer** and **denatures membrane proteins**; measure with a **colorimeter** and state the filter.

CALCULATIONS

- **% change** = $(\text{new} - \text{original}) \div \text{original} \times 100$. Keep the sign; a loss is negative.
- **Rate** = change $\div$ time, **with units** (write $\text{mm}^3 \text{min}^{-1}$, not mm^3/min).
- **Magnification** = image $\div$ actual. **Convert to the same unit first:** 1 mm = 1000 μm .
- **Mean** to the same number of decimal places as the raw data.
- **Standard deviation** measures the **spread** about the mean. A **larger** SD means more variable data.

INTERPRETING ERROR BARS — THE ONLY SIGNIFICANCE WORK REQUIRED

Overlap → the difference is not significant. **No overlap** → it is significant. **Always name the two groups you are comparing** — “overlapping standard deviations are not significant” unqualified is not accepted. Add that **a statistical test would be needed to confirm it**.

TABLES

Three marks are usually available: a **suitable format with the raw data**; **headings with units, the unit in the heading only** (units in the body are not accepted); and **means given to the same number of decimal places as the data**. **An extra column negates the first mark** — no “trial number” column, and no calculated rate inside a raw-data table.

EVALUATION AND IMPROVEMENT

Weakness	Improvement that scores
Too few readings	Take at least five values of the IV across a wider range
No repeats	Repeat each at least three times and take a mean
Subjective judgement (colour, end point)	Use a colorimeter ; or judge against a standard; or have the same person judge throughout
Imprecise measurement	Use an instrument of finer resolution ; measure a larger quantity so the percentage uncertainty falls
Uncontrolled variable	Name it and say how to control it

Anomalies: identify the point, say it does **not fit the trend**, suggest a cause, and say it should be **repeated** — not simply deleted.

Risk is marked as a **named hazard plus a matching reduction**. “Wear a lab coat” and “be careful” score nothing. Scalpel or cork borer: cut on a tile away from the fingers. Microbes: aseptic technique, tape the lid, incubate below 30 °C. Hot water baths: use tongs.

DRAWING FROM A MICROSCOPE

Use a **sharp pencil**, **clear unbroken lines**, **no shading**, drawn to a reasonable size. Label lines must be **straight, drawn with a ruler, not crossing**, and must **touch the structure**. Give a **title** and a **magnification or scale bar**. Draw **only what you can see** — a plan drawing shows tissue outlines, not individual cells.

THE REJECTIONS BANK

Do not write	Write instead
“Amount”	Volume, mass or number , with a unit
“Different concentrations”	Stated values: 0.1, 0.2, 0.3, 0.4 and 0.5 mol dm^{-3}
“Several readings”	At least five , each repeated
“It would affect the results”	A directional effect on the DV
“Water bath” unqualified	Thermostatic water bath
“Repeating removes random error”	It reduces the effect of random error
“Not reliable” / “not accurate”	The results would not be valid
Units in the body of a table; a trial-number column	Units in the heading only ; no extra column
Dot-to-dot on a scatter graph	A single thin best-fit line
“The SDs overlap so it is not significant”	Name the two groups being compared
“Be careful” / “wear goggles” alone	A named hazard with a matching reduction

WALKING INTO THE EXAM

You have 1.6 minutes per mark — more than any other Biology paper. Use it to write **every** method point rather than a tidy paragraph. State numbers. Put units in table headings and on graph axes. Make every effect directional, and name the groups whenever you comment on error bars.